



Database Design & Implementation CSCI-UA060

Normalization Workshop



Exercise 1 - Demographics

demographics

Country	Capital	EthnicGroup	Percentage	Population	Language1	Language2
Freeland	Utoptown	Fens	34	5634476	Scrumish	English
Freeland	Utoptown	Junts	66	5634476	Scrumish	English
Othermany	Greattown	Sorns	46	43526	English	Otherese
Othermany	Greattown	Fens	54	43526	English	Otherese
Summerland	Sun City	Shades	30	663524	Mandarin	Orange
Summerland	Sun City	Brights	70	663524	Mandarin	Orange
Snowdonia	Big Flake	Sorns	100	6344335	Fernish	Scrumish

Does this table have a primary key? If so, what is it?

Is this table in a Normal Form? If so, what Normal Form is it in?

How would you set up this data in a database?

What assumptions, if any, are you making?

Exercise 2 - Room

room										
roomName	size	wallColor	paintType	paintCompany	paintSKU	AcOutlets	numberWindows	windowExposure	flooring	floorColor
Living Room	300	light green	satin	Benjamin Moore	33-663-23	4	1	East	Carpet	Green
Kitchen	150	Yellow	gloss	Behr	34-662-34	5	1	North	Tile	White
Bedroom	200	Blue	Matte	Benjamin Moore	42-746-54	3	2	East, South	Carpet	Blue
Bedroom 2	150	Grey	Matte	Behr	64-223-55	4	2	South, West	Wood	Brown

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Exercise 3 – Order Detail

order_detail

OrderId	OrderLineNumber	CustomerId	CustomerName	ProductName	Color	Manufacturer	UnitCost	Quantity	TotalCost
10354	1	3564	Smith	Ceiling Fan	White	Hampton Bay	354	1	354
10357	1	5643	Brown	Ceiling Fan	Black	Hampton Bay	365	2	730
10357	2	5643	Brown	Sconce	Bronze	Hunter	276	1	276
10360	1	3564	Smith	Chandelier	Gold	Monte Carlo	434	1	434
10361	1	4563	Jones	Sconce	Nickel	Hampton Bay	344	3	1032
10361	2	4563	Jones	Chandelier	Silver	Hunter	445	1	445
10362	1	4475	Wilson	Lamp	Brown	Progress	654	1	654

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Exercise 4 – Rangers Scores

scores

date	team	result	score	location	teamCaptain
1-Sep	Islanders	Win	3-1	Home	Lee
3-Sep	Red Wings	Loss	2-4	Home	Howard
4-Sep	Penguins	Win	3-2	Away	Crosby
5-Sep	Islanders	Loss	1-2	Away	Lee
7-Sep	Blackhawks	Loss	1-5	Away	Kane
10-Sep	Bruins	Win	4-1	Home	Rask

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Exercise 5 – Baseball Scores

scores

date	homeTeam	visitingTeam	score	numberHomeRuns	homeTeamManager
1-Sep	Yankees	Dodgers	6-3	2	Boone
1-Sep	Red Sox	Cubs	3-4	1	Roenicke
4-Sep	Yankees	Braves	3-8	0	Boone
4-Sep	Blue Jays	Cardinals	16-2	5	Montoyo
4-Sep	Cardinals	Cubs	10-11	4	Shildt
10-Sep	Blue Jays	Red Sox	1-0	1	Montoyo

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Exercise 6 – Car Rental

rental						
customerId	carId	carColor	carType	dateOut	datein	agentId
34432	432	Red	Sedan	9/1	9/2	43
36574	354	Red	SUV	9/3	9/10	23
35432	432	Red	Sedan	9/5	9/6	34
36475	446	Black	Coupe	9/6	9/15	23
36465	324	White	Sedan	9/7	9/8	32

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